Sequence-to-Sequence (Seq2Seq) Models

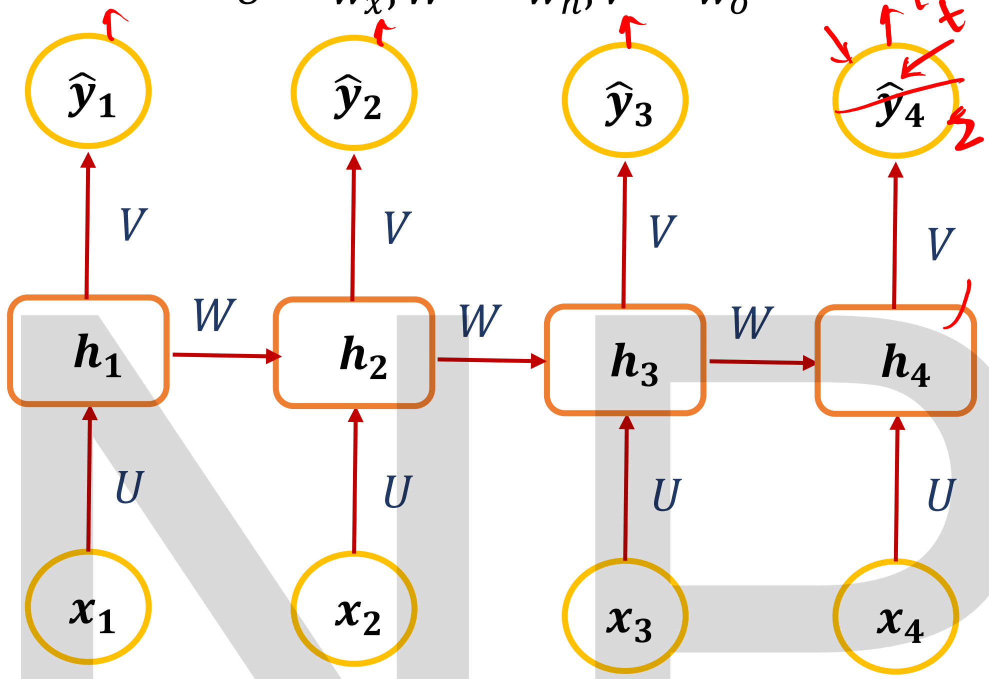
Previous session

- Derivative of sigmoid
- Vanishing gradient and Exploding gradient problem in Deep NN

Today's Session Overview

- Backpropagation Through Time (BTT)
- Challenges such as Vanishing and Exploding Gradients in RNN
- How to fix exploding gradients

$U = w_x, W = w_h, V = w_o$



$V_{new} = V_{old} - \eta \frac{\partial L}{\partial V}$

$V \dashrightarrow z_t \dashrightarrow \hat{y}_t \dashrightarrow L$

$$\left(\frac{\partial L}{\partial V}\right)_t = \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial V}$$

$$\frac{\partial L}{\partial V} = \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial V}$$

$$= (\hat{y}_t - y_t) \cdot 1 \cdot h_t$$

$z_t = h_t \cdot V + b_0$

$\frac{\partial z_t}{\partial V} =$

sigmoid

linear

$\hat{y}_t = f(z_t)$

$f(x) = x$

$\hat{y}_t = z_t$

$\frac{\partial \hat{y}_t}{\partial z_t}$

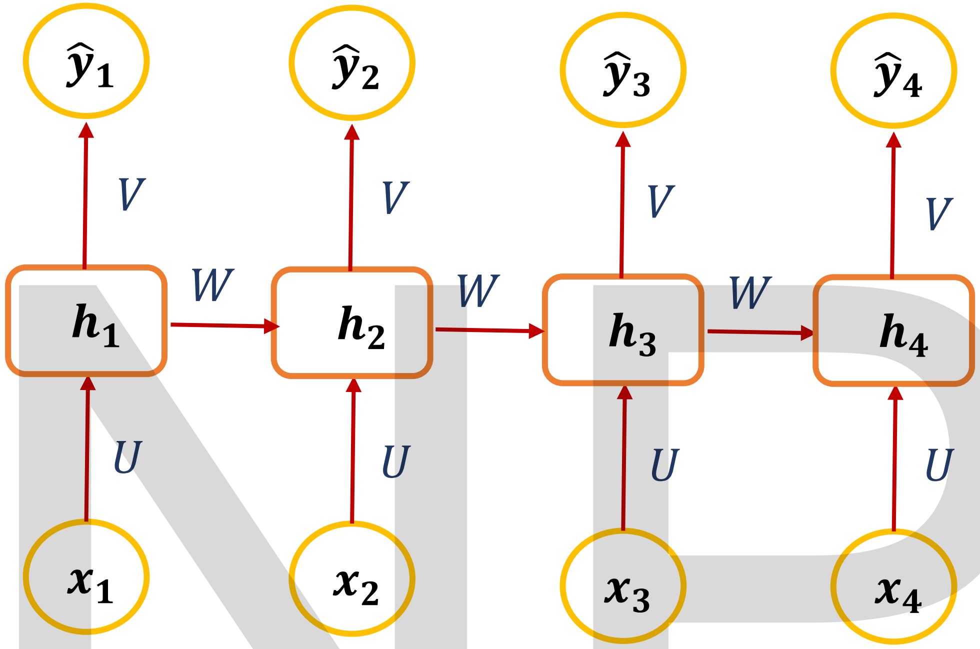
$$L = \frac{1}{2}(y_t - \hat{y}_t)^2$$

$$= \frac{1}{2} * 2 (y_t - \hat{y}_t)(-1)$$

$$= -(y_t - \hat{y}_t)$$

$$= -y_t + \hat{y}_t \rightarrow (\hat{y}_t - y_t)$$

$$U = w_x, W = w_h, V = w_o$$



At a given time step t:

Pre-activation (logit):

$$z_t = Vh_t + b_0$$

Output:

•If **linear output** (regression):

$$\hat{y}_t = z_t$$

•If **sigmoid output** (binary classification):

$$\hat{y}_t = \sigma(z_t)$$

Loss (MSE):

$$L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2$$

At time step t, contribution to the gradient of V:

$$\frac{\partial L_t}{\partial V} = \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial V}$$

$$V \dashrightarrow z_t \dashrightarrow \hat{y}_t \dashrightarrow L$$

$$\frac{\partial L}{\partial V} = \sum_{t=1}^{T} \frac{\partial L}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial V}$$

$$V_{new} = V_{old} - \eta \frac{\partial L}{\partial V}$$

Linear Output Layer
(Most common in RNN regression)

$$L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2 \qquad \frac{\partial L_t}{\partial \hat{y}_t} = \hat{y}_t - y_t$$

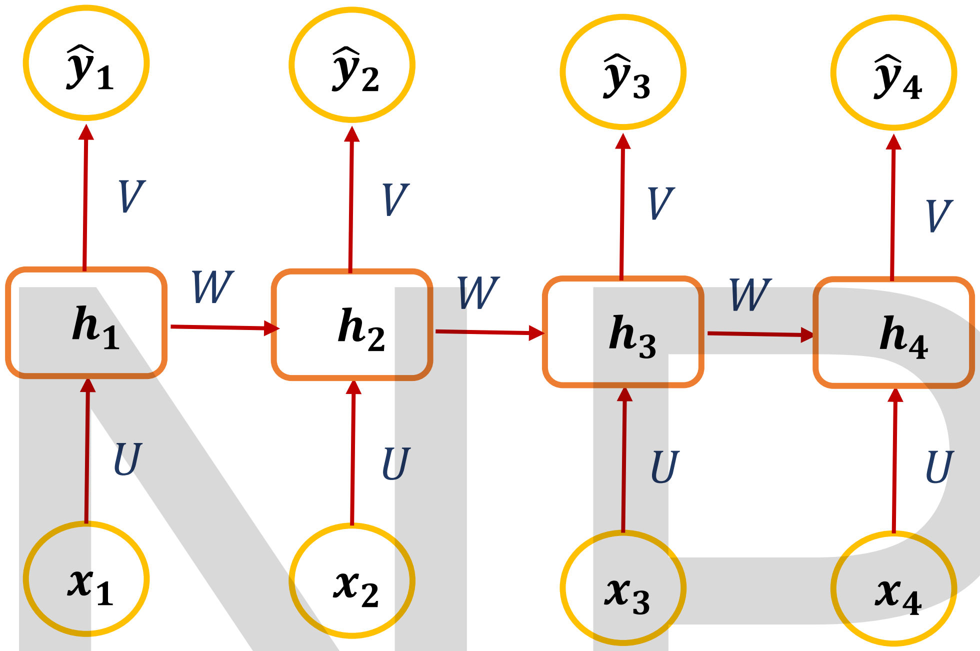
$$\frac{\partial \hat{y}_t}{\partial z_t} \qquad \text{Since: } \hat{y}_t = z_t \Rightarrow \frac{\partial \hat{y}_t}{\partial z_t} = 1$$

$$\frac{\partial z_t}{\partial V} \qquad z_t = Vh_t + b_0 \Rightarrow \frac{\partial z_t}{\partial V} = h_t$$

$$\frac{\partial L_t}{\partial V} = (\hat{y}_t - y_t) \cdot 1 \cdot h_t = (\hat{y}_t - y_t) h_t \qquad \frac{\partial L}{\partial V} = \sum_{t=1}^{T} (\hat{y}_t - y_t) \cdot h_t$$

If hidden state h_t is large → output weight update is large
If prediction error is large → update is large

$U = w_x, W = w_h, V = w_o$



$$\delta_1 = g_1 + g_2 f_2 + g_3 f_3 f_2 + g_4 \cdots$$

$$\delta_1 = \frac{\partial L_1}{\partial h_1} + \frac{\partial L_2}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1} + \frac{\partial L_3}{\partial h_3} \cdot \frac{\partial h_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1}$$

$$\boxed{\delta_t = \sum_{k=t}^{T} \left(\frac{\partial L_k}{\partial h_k} \prod_{j=t+1}^{k} \frac{\partial h_j}{\partial h_{j-1}} \right)}$$

$$\frac{\partial h_{t+1}}{\partial h_{t+x-1}} = \frac{\partial h_{t+1}}{\partial h_t}$$

→ vanishing

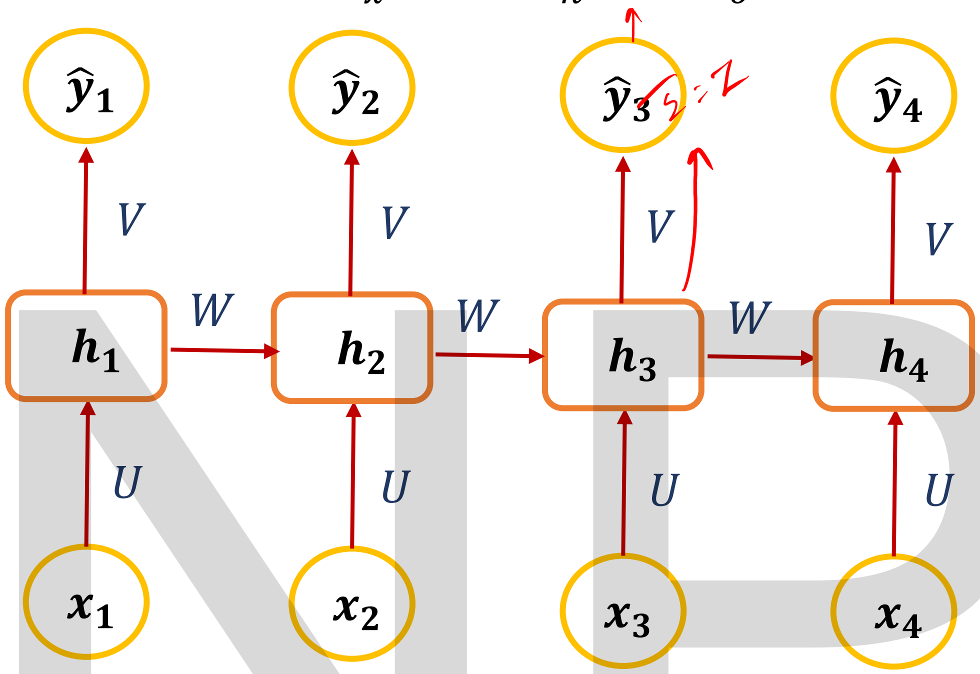
exploding

$$\delta_t = \sum_{k=t}^{T} (\text{contribution of } L_k \text{ to } h_t)$$

$k=t$ $\quad$ $k=t+1 \rightarrow$

$k=t+2$

$$U = w_x, W = w_h, V = w_o$$



At time step t:

- Hidden pre-activation: $a_t = Ux_t + Wh_{t-1} + b_x$
- Hidden state: $h_t = \tanh(a_t)$
- Output: $\hat{y}_t = f(Vh_t + b_o), \textit{linear activation}$
- Loss per time step (MSE): $L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2$
- Total loss:

$$L = \sum_{t=1}^{T} L_t$$

At time step t, contribution to the gradient of W: $\left(\frac{\partial L}{\partial W}\right)_t = \frac{\partial L_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial a_t} \cdot \frac{\partial a_t}{\partial W}$

$$\frac{\partial L}{\partial W} = \sum_{t=1}^{T} \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial a_t} \cdot \frac{\partial a_t}{\partial W}$$

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W}$$

$$\frac{\partial a_t}{\partial W} = h_{t-1} \qquad a_t = x_t U + h_{t-1} W + b_x$$

$$\frac{\partial h_t}{\partial a_t} = 1 - h_t^2$$

Activation function used is tanh. Derivative of tanh is

$$1 - tanh^2(x)$$

$$\tanh(a_t) = h_t$$

$$tanh'(a_t) = 1 - h_t^2$$

$$\left(\frac{\partial L}{\partial W}\right)_t = \left(\frac{\partial L}{\partial h_t}\right) \cdot 1 - h_t^2 \cdot h_{t-1}$$

h_t goes **up** to $\hat{y}_t$ through V

h_t goes **right** to h_{t+1} through W

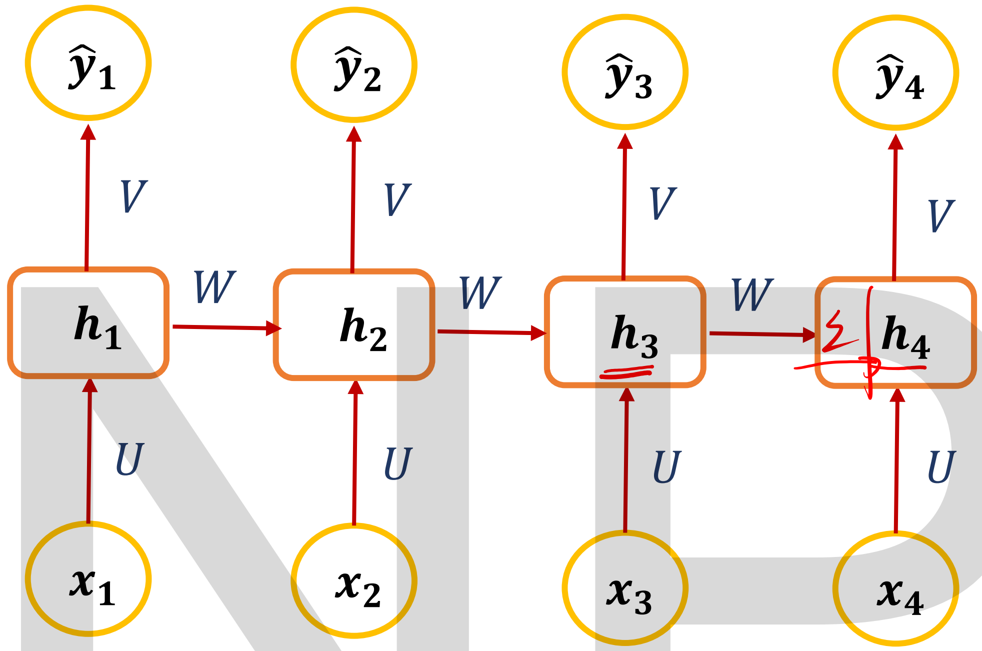
$$h_t \rightarrow z_t \rightarrow \hat{y}_t \rightarrow L_t$$

$$\frac{\partial L_t}{\partial h_t} = \underbrace{\frac{\partial L_t}{\partial \hat{y}_t}}_{\hat{y}_t - y_t} \cdot \underbrace{\frac{\partial \hat{y}_t}{\partial z_t}}_{1} \cdot \underbrace{\frac{\partial z_t}{\partial h_t}}_{V}$$

$$z_t = h_t V + b_o$$

$$\boxed{\frac{\partial L_t}{\partial h_t} = (\hat{y}_t - y_t) \cdot V}$$

$U = w_x, W = w_h, V = w_o$



At time step t:

- Hidden pre-activation: $a_t = Ux_t + Wh_{t-1} + b_x$
- Hidden state: $h_t = \tanh(a_t)$
- Output: $\hat{y}_t = f(Vh_t + b_o)$, *linear activation*
- Loss per time step (MSE): $L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2$
- Total loss:

$$L = \sum_{t=1}^{T} L_t$$

$$\left(\frac{\partial L}{\partial W}\right)_t = \left(\frac{\partial L}{\partial h_t}\right) \cdot 1 - h_t^2 \cdot h_{t-1}$$

h_t goes **up** to $\hat{y}_t$ through V

h_t goes **right** to h_{t+1} through W

Say $h_3 \rightarrow h_4 \rightarrow L_t$

$\boxed{h_t \rightarrow h_{t+1}} \rightarrow L \rightarrow \frac{\partial L}{\partial h_{t+1}} \cdot \frac{\partial h_{t+1}}{\partial h_t}$

$h_t \rightarrow a_{t+1} \rightarrow h_{t+1}$

$$\frac{\partial h_{t+1}}{\partial h_t} = \underbrace{\frac{\partial h_{t+1}}{\partial a_{t+1}}}_{(1 - h_{t+1}^2)} \cdot \underbrace{\frac{\partial a_{t+1}}{\partial h_t}}_{W}$$

$$a_{t+1} = x_{t+1}U + h_tW + b_x$$
$$h_{t+1} = \tanh(a_{t+1})$$
$$= (1 - h_{t+1}^2)$$

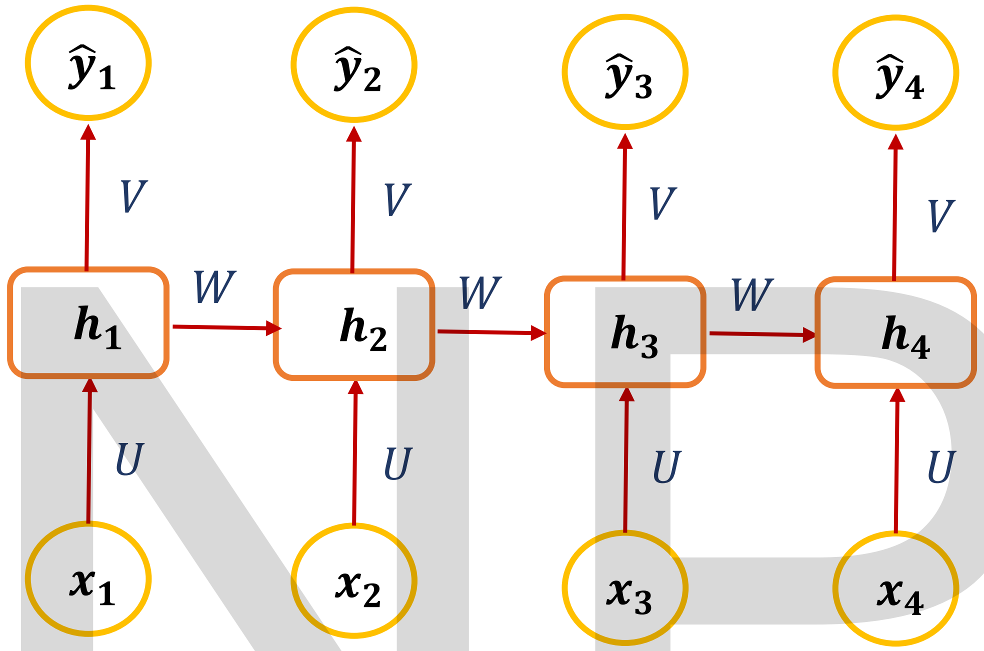
So $\frac{\partial L}{\partial h_t}$ affects:

1. the **current loss** L_t through $\hat{y}_t$, and
2. all **future losses** L_{t+1}, L_{t+2}, L_T through h_{t+1}, h_{t+2}, h_T.

$$\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + \frac{\partial L}{\partial h_{t+1}} \cdot \frac{\partial h_{t+1}}{\partial h_t}$$

exact BPTT definition

$$U = w_x, W = w_h, V = w_o$$



At time step t:

- Hidden pre-activation: $a_t = Ux_t + Wh_{t-1} + b_x$
- Hidden state: $h_t = \tanh(a_t)$
- Output: $\hat{y}_t = f(Vh_t + b_o), \textit{linear activation}$
- Loss per time step (MSE): $L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2$
- Total loss:

$$L = \sum_{t=1}^{T} L_t$$

$$\underbrace{\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t}}_{(\hat{y}_t - y_t)\cdot V} + \underbrace{\frac{\partial L}{\partial h_{t+1}} \cdot \frac{\partial h_{t+1}}{\partial h_t}}_{\frac{\partial L}{\partial h_{t+1}} \cdot (1 - h_{t+1}^2) \cdot W}$$

exact BPTT definition

$$\frac{\partial L}{\partial h_t} = (\hat{y}_t - y_t) \cdot V + \frac{\partial L}{\partial h_{t+1}} \cdot (1 - h_{t+1}^2) \cdot W$$

$$\frac{\partial L}{\partial W} = \sum_{t=1}^{T} (\hat{y}_t - y_t) \cdot V + \frac{\partial L}{\partial h_{t+1}} \cdot (1 - h_{t+1}^2) \cdot W \cdot 1 - h_t^2 \cdot h_{t-1}$$

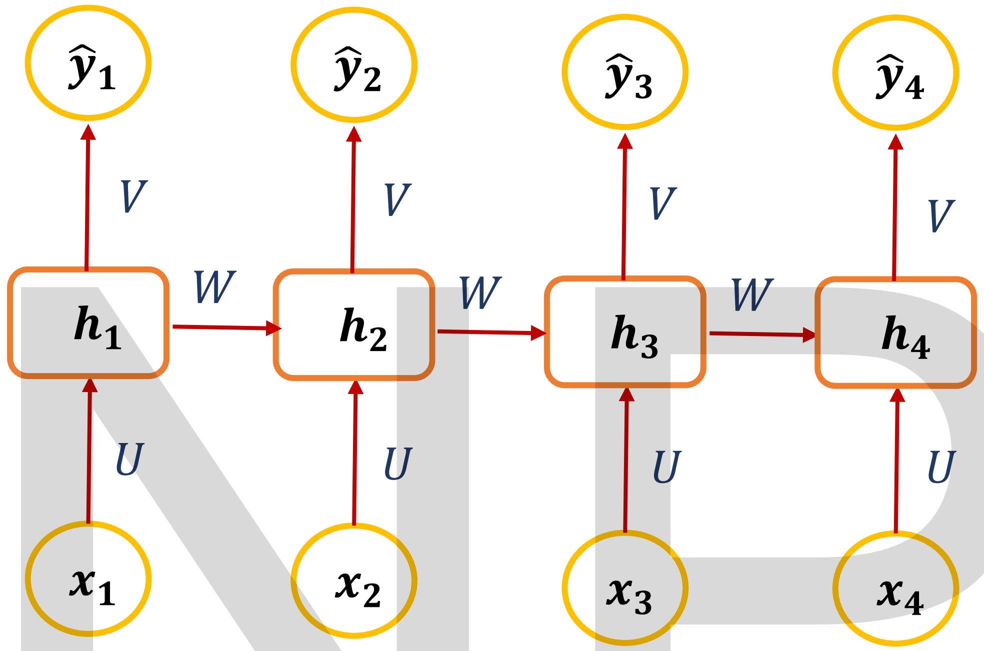
$$\underbrace{\frac{\partial L}{\partial h_t}}_{\delta_t} = \underbrace{\frac{\partial L_t}{\partial h_t}}_{g_t} + \frac{\partial L}{\partial h_{t+1}} \cdot \underbrace{\frac{\partial h_{t+1}}{\partial h_t}}_{A_{t+1}}$$

$$\delta_t = g_t + \frac{\partial L}{\partial h_{t+1}} \cdot A_{t+1}$$

$$\delta_t = g_t + \delta_{t+1} \cdot A_{t+1}$$

At the final step:
$\delta_T = g_T$

$U = w_x, W = w_h, V = w_o$



At time step t:

•Hidden pre-activation: $a_t = Ux_t + Wh_{t-1} + b_x$

•Hidden state: $h_t = \tanh(a_t)$

•Output: $\hat{y}_t = f(Vh_t + b_o), \textit{linear activation}$

•Loss per time step (MSE): $L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2$

•Total loss:

$$L = \sum_{t=1}^{T} L_t$$

$$\delta_t = g_t + \delta_{t+1} \cdot A_{t+1}$$

At the final step:
$\delta_T = g_T$

Take a tiny sequence length $T = 3$(just for pattern):

At $t = 3 \coloneqq \delta_3 = g_3$

At $t = 2 \coloneqq \delta_2 = g_2 + \delta_3 A_3 = g_2 + g_3 A_3$

At $t = 1$

$$\begin{aligned} \delta_1 &= g_1 + \delta_2 A_2 \\ &= g_1 + (g_2 + g_3 A_3) A_2 \\ &= g_1 + g_2 A_2 + g_3 A_3 A_2 \end{aligned}$$

$$\delta_1 = g_1 + g_2 A_2 + g_3 A_2 A_3$$

$$\delta_1 = \frac{\partial L_1}{\partial h_1} + \frac{\partial L_2}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1} + \frac{\partial L_3}{\partial h_3} \cdot \frac{\partial h_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1}$$

So concretely:

$$A_2 = \frac{\partial h_2}{\partial h_1}$$

$$A_3 = \frac{\partial h_3}{\partial h_2}$$

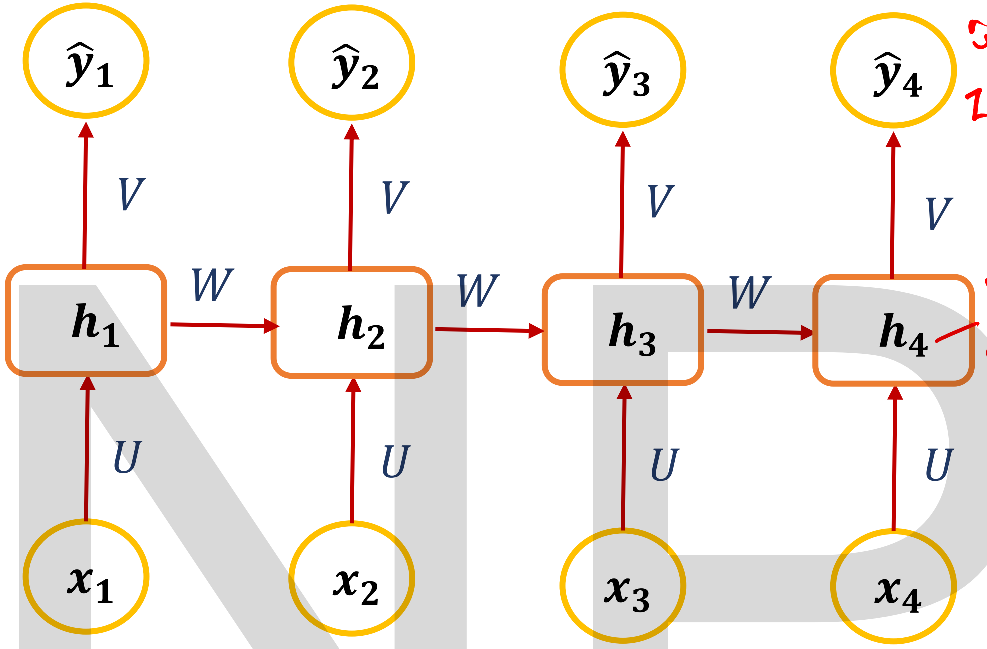
$$\delta_t = \sum_{k=t}^{T} \left(\frac{\partial L_k}{\partial h_k} \prod_{j=t+1}^{k} \frac{\partial h_j}{\partial h_{j-1}} \right)$$

Vanishing and exploding gradients come ONLY from this product term

$$\frac{\partial L}{\partial W} = \sum_{t=1}^{T} \delta_t \cdot 1 - h_t^2 \cdot h_{t-1}$$

$$\delta_t = (\hat{y}_t - y_t)V + \delta_{t+1}(1 - h_{t+1}^2)W$$

$U = w_x, W = w_h, V = w_o$



$U \longrightarrow a_t \longrightarrow h_t \longrightarrow z_t \rightarrow \hat{y}_t \rightarrow L_t$

$$\left(\frac{\partial L}{\partial U}\right)_t \cdot \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial a_t} \cdot \frac{\partial a_t}{\partial U}$$

$$\left(\frac{\partial L}{\partial U}\right)_t = (\hat{y}_t - y_t) \cdot 1 \cdot V \, (1-h_t^2) \cdot x_t$$

$$a_t = x_t \cdot U + h_{t-1} \cdot W + b_a$$

$$h_t = \tanh(a_t)$$

$$= 1 - h_t^2$$

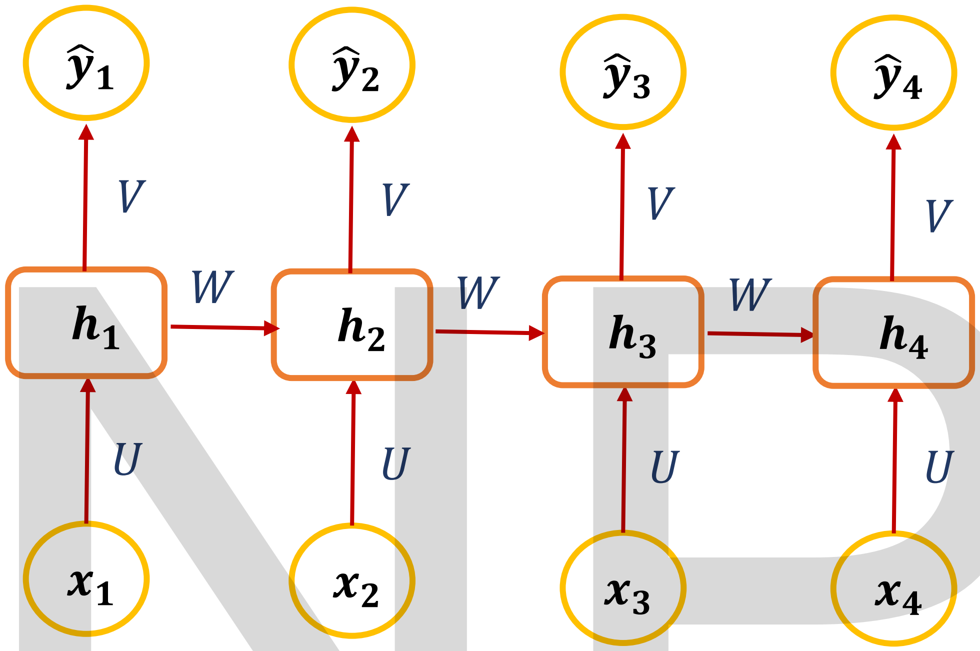
$$\frac{\partial L}{\partial U} = \sum_{t=1}^{T} \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial a_t} \cdot \frac{\partial h_t}{\partial U}$$

$$\boxed{\frac{\partial L}{\partial U} = \sum_{t=1}^{T} \delta_t \cdot (1-h_t^2) \cdot x_t}$$

$$z_t = h_t \cdot V + b_o$$

$$\delta_t = (\hat{y}_t - y_t) \cdot V + \delta_{t+1} \cdot (1-h_{t+1}^2) \cdot W$$

$$U = w_x, W = w_h, V = w_o$$



At time step t:

- Hidden pre-activation: $a_t = Ux_t + Wh_{t-1} + b_x$
- Hidden state: $h_t = \tanh(a_t)$
- Output: $\hat{y}_t = f(Vh_t + b_o)$, *linear activation*
- Loss per time step (MSE): $L_t = \frac{1}{2}\left(y_t - \hat{y}_t\right)^2$
- Total loss:

$$L = \sum_{t=1}^{T} L_t$$

$$\frac{\partial L}{\partial U} = ?$$

$$U_{new} = U_{old} - \eta \frac{\partial L}{\partial U}$$

Which path connects U to the loss?

Look at one time step t:

$$U \longrightarrow a_t \longrightarrow h_t \longrightarrow z_t \rightarrow \hat{y}_t \rightarrow L_t$$

So the chain rule for that time step is:

$$\left(\frac{\partial L}{\partial U}\right)_t = \frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial a_t} \cdot \frac{\partial a_t}{\partial U}$$

$$\frac{\partial L_t}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial z_t} \cdot \frac{\partial z_t}{\partial h_t} = (\hat{y}_t - y_t) \cdot 1 \cdot V \qquad \frac{\partial h_t}{\partial a_t} = (1 - h_t^2) \qquad \frac{\partial a_t}{\partial U} = x_t$$

$$a_t = Ux_t + Wh_{t-1} + b_x$$

$$\left(\frac{\partial L_t}{\partial U}\right)_t = (\hat{y}_t - y_t) \cdot 1 \cdot V \cdot (1 - h_t^2) \cdot x_t$$

Then we sum these contributions over all time steps $t = 1, \ldots, T$:

$$\frac{\partial L}{\partial U} = \sum_{t=1}^{T} \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial a_t} \cdot \frac{\partial a_t}{\partial U} \qquad \frac{\partial L}{\partial U} = \sum_{t=1}^{T} \delta_t \cdot (1 - h_t^2) \cdot x_t \qquad \delta_t = \frac{\partial L}{\partial h_t}$$

$$\delta_t = (\hat{y}_t - y_t)V + \delta_{t+1}(1 - h_{t+1}^2)W$$

How to fix exploding gradients

→ in general

1. Gradient Clipping (MOST IMPORTANT – especially for RNNs):

Limit the magnitude of gradients before updating weights.

Stage 1: When training starts, we already have:

Given (Fixed): Input data x and Target y

Initialized (Model's starting point)

- Weights $w_1, w_2, \ldots$
- Biases b

These initial weights are randomly initialized (small values).

Stage 3: Backpropagation

Mathematically:

$$\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial b}$$

These derivatives are called **gradients**.

Example:

$$\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial b} = [12, 16, 9]$$

Step 1: Compute gradient norm

$$\| g \|_2 = \sqrt{12^2 + 16^2 + 9^2} = \sqrt{144 + 256 + 81} = \sqrt{481} \approx 21.93$$

Stage 2: Forward Pass (No gradients yet)

Using the current weights and bias:

$$z = wx + b$$

$$\hat{y} = f(z)$$

Then we compute loss:

$$L = Loss(y, \ \hat{y})$$

Step 2: Choose clipping threshold

Let the clipping threshold be:

$$\tau = 5$$

Since $\| g \|_2 = 21.93 > 5$,we clip.

Step 3: Compute scaling factor

$$\alpha = \frac{\tau}{\| g \|_2} = \frac{5}{21.93} \approx 0.228$$

Step 4: Scale the gradient

$$g_{clipped} = \alpha \cdot g = 0.228 \cdot [12, 16, 9] \approx [2.736, 3.648, 2.052]$$

Let learning rate $\eta = 0.1$, and current parameters:

$$\theta = [w_1, w_2, b] = [1.0, -2.0, 0.5]$$

Update rule:

$$\theta_{new} = \theta - \eta g_{clipped}$$

So:

- $w_{1(new)} = 1.0 - 0.1(2.736) = 1.0 - 0.2736 = 0.7264$
- $w_{2(new)} = -2.0 - 0.1(3.648) = -2.0 - 0.3648 = -2.3648$
- $b_{new} = 0.5 - 0.1(2.052) = 0.5 - 0.2052 = 0.2948$

2. Proper Weight Initialization:

Start training with reasonable weight magnitudes.

Network Type	Initialization
Deep NN (ReLU)	He Initialization
Deep NN (tanh/sigmoid)	Xavier / Glorot
RNN	Orthogonal initialization (for recurrent weights)

3. Use Appropriate Activation Functions

Use ReLU & Variants: Rectified Linear Units (ReLU) and their variants (Leaky ReLU) prevent saturation and large gradients better than sigmoid/tanh

4. Regularization & Optimizers:

L2 Regularization, Optimizers like RMSprop or Adam

5. Dropout

6. Batch normalization

7. Learning rate control:

Gradient descent update rule:

$$w_{new} = w_{old} - \eta \cdot \frac{\partial L}{\partial w}$$

Two extreme cases: Learning rate too small

- Weight updates are tiny
- Training is very slow
- Model may never converge (within reasonable time)

Learning rate too large

- Weight updates overshoot minimum
- Loss oscillates or diverges
- Can cause exploding gradients

So, adjust the learning rate during training.

Next Session

Hands-on: Implementation of RNN using Keras (Time-series example)

Thank you